Using SMS to Transfer Small Data Packets During Periods of High Workload on Mobile Data Networks

Hayala N. Curto[1], Josemar Caetano[1], Jussara Almeida[2], Artur Ziviani[3], Carlos Henrique S. Malab[4], Humberto T. Marques-Neto[1]

[1]Department of Computer Science
Pontifical Catholic University of Minas Gerais (PUC Minas)
Belo Horizonte – MG – Brazil – 30.535-901

[2]Department of Computer Science
Federal University of Minas Gerais (UFMG)
Belo Horizonte, MG – 31.270-010

[3]National Laboratory for Scientific Computing
(LNCC/MCTI) – Petropolis, RJ – Brazil – 25.651-075

[4]Oi Telecom
Rio de Janeiro, RJ 20.230-070

{hayala.curto,josemar.caetano}@sga.pucminas.br, jussara@dcc.ufmg.br,

ziviani@lncc.br, malab@oi.net.br, humberto@pucminas.br

Abstract. *This paper proposes an SMS-based platform for transferring small data packets, allowing mobile applications to operate even in situations in which the data traffic over mobile data networks is unavailable or under congestion. We first conduct a workload characterization of a Brazilian mobile network during recent large-scale events to verify calls and SMS usage patterns at such events. This characterization pointed out that voice calls and SMS work relatively well in these large-scale events, despite frequent reports from users about the data network congestion. As both transmission and reception of SMSes tend to remain in operation even during network overload, we propose in this paper the usage of SMSes to transfer small data packets during such periods. Thus we will keep key mobile applications transferring data even during periods of high workload or unavailability of the mobile network. We believe that this work could improve users' experience during large scale events and similar situations where mobile data networks are under congestion or unavailable.*

1. Introduction

The usage of mobile applications during large-scale events, such as concerts, conferences, or sport matches, has increased considerably in recent years. Typical attendees look for the most enriching experience using the applications running on their mobile devices. However, often the mobile network in the surroundings of these events can be overloaded, it may frustate a large number of people. On the other hand, despite the overload of data network, mobile operators can still keep the network calls and SMS traffic operating with certain availability.

We note however that, during periods of network congestion, SMS traffic tends to be the last one to become unavailable. Thus, in order to improve user experience, in this paper we present a new platform with an Application Programming Interface (API) to support data traffic through SMS. This platform could improve the value of mobile application for users. Towards designing such platform, we first characterize the workload of a major mobile phone network on different days when major events (notably, major soccer matches) took place in Rio de Janeiro, Brazil. We analyze the usage behavior of three profiles of attendees of two major matches of the FIFA 2014 World Cup, including the final between Germany and Argentina on July 13, 2014. We compare these profiles with similar results computed on a dataset of calls and messages collected on the day of the final match of 2013 Brazilian Cup, which happened at the same stadium. Each profile consists of: (i) attendees who made cellphone calls and sent SMS messages, (ii) attendees who only made calls, and (iii) attendees who only sent messages.

Users in each profile tend to have different behaviors in different periods of the event. Our characterization revealed that voice calls and SMS are still in use in these large-scale events, despite frequent reports from users about the unavailability of the data network. There is always a physical limit of the available network at large-scale events. Therefore, mobile operators typically also use a Wi-Fi network in the most relevant events [Estadao 2013].

This characterization was done with anonymized datasets of cellphone calls and SMS messages, provided by a major Brazilian cellphone carrier. The dataset does not include any trace from data network. We use the position of cellular operator's base stations to get approximate location of the user. Based on the results from this characterization, we propose a platform that uses a sequence of SMSes to exchange data between any Android mobile application and web services. Allow applications to transfer small data packets is the main contribution of this work.

Moreover, Brazilian mobile operators recently announced their intent to work on changing their mobile data contracts in order to consider completely blocking the user data access (rather than only slowing down the transmission rate) once the monthly data amount contracted by user is ended. On the other hand, the offer of unlimited SMS messages is growing as marketing actions of these operators, thus motivating the design of solutions, such as our proposed platform, which can exploit them to improve user experiences with their mobile devices.

The paper is organized as follows. Section 2 discusses the related work. Section 3 presents the methodology to characterize voice calls and SMS in large-scale events. The characterization results are shown in Section 4. Section 5 introduces the proposed SMS-based platform for transferring small data packages. Section 6 presents the usage of this proposed platform. Final remarks are discussed in Section 7.

2. Related Work

[Oliver 2010] and [Chen et al. 2009] characterize SMS user behavior and present some advantages of texting as a communication tool. One of them is the worldwide availability of this technology. The SMS is well-known and well-used in both developed and developing countries, and uses a communication standard with penetration in more than 220 countries. The asynchronous nature of SMS is another positive aspect, as well as its low cost when compared with other forms of human communication [Fjeldsoe et al. 2009]. [Oliver 2010] also build an efficient and reliable data transport protocol on top of the Short Message Service (SMS). Our SMS-based platform uses and extends this work, including a mobile phone as a gateway to allow data traffic in SMS.

The massive adoption of mobile devices has motivated researchers of different areas to study economic and behavioral aspects related to the use of such devices [Eagle and Pentland 2006, Eagle and Pentland 2009], as well as human mobility patterns [Lee et al. 2009, Candia et al. 2008, Gonzalez et al. 2008, Soper 2012, Becker et al. 2013] are inferred from it. For instance, [Ratti et al. 2006] reviewed the potential of location-based services (LBS), using mobile phones, to the urban planning community, with the developing of a graphic representation of the intensity of urban activities and their evolution through space and time.

In [Xavier et al. 2013], authors presented a methodology to analyze the workload dynamics of a cellular network during large-scale events with a specific focus on soccer matches. We extend his work, exploiting also SMS messages, in addition to cellphone calls, and by identifying usage profiles. In Section 3, we detail these new features.

The SMS usage as an alternative in environments with low availability of network resources was also proposed by [Veeraraghavan et al. 2007] and [Reda et al. 2010]. In the first paper [Veeraraghavan et al. 2007], authors presented a low-cost solution for small amounts of important information traffic, replacing PC-based systems by phone-based systems. In the second one [Reda et al. 2010], authors proposed a framework for distributing individualized personal data in environments with low availability of network resources. Experimental results showed significant time saving for the end user, through prefetching scheduled personal data. The 3G networks unavaiability was described by [Zhuo et al. 2011]. The work of [Cheung and Huang 2013] proposes the usage of Wi-Fi networks to help the mobile operator to obtain immediate capacity relief when facing the explosive growth of mobile data traffic in 3G networks. A preliminary version of this work was recently published as a 2-page extended abstract [Curto et al. 2014].

3. Methodology to characterize voice calls and SMS in large-scale events

In [Xavier et al. 2013], the authors proposed a methodology to characterize voice calls during large-scale events that aims at improving the planning and management of the cellphone network resources, specially in order to accommodate the large demands that arise during large-scale events. One of its key step is the identification of cellphone users involved with a large-scale event, which enabled the characterization of workload imposed on the carrier's infrastructure. In that paper, the dataset used to validate the proposed methodology included only voice calls made by cellphone users of a major Brazilian carrier, who are completely anonymized with an unique identification. That dataset has also information about the carrier's base stations (i.e., the geographical locations) in which each voice call started and finished. In the present work, we extend this methodology in order to improve the analyses made in [Xavier et al. 2013] by also exploiting data about SMS messages (in addition to phone calls) and by analyzing the behavior of three usage profiles. In this section, we detail these new features added to the previous methodology.

As in prior work, the main assumption behind the proposed methodology is that by identifying people who made calls and sent SMS messages through the base stations covering the region of the event location around the time the event happened, we can focus on the group of users involved directly or indirectly with the analyzed event, here referred to *attendees*[1]. Our main goal is also to identify those attendees and understand their behavior, in terms of network usage, before and after the event. To this end, we also use completely anonymized datasets with cellphone calls and SMS messages collected at a carrier infrastructure (i.e., at its base stations) that cover both the location of the selected events and its main access routes. The proposed methodology is divided into five main steps, which are described in the following sections.

3.1. Step 1 – Defining User Profiles

Different users may exhibit different profiles in terms of the types of mobile services they most often use to communicate during large-scale events. Since different services impose different loads on the carrier's infrastructure, we choose to add an initial step in the methodology which consists of defining the user profiles.

Considering the two main types of mobile services available (voice calls and SMS messages), we envision three distinct user profiles. One profile consists of users who only make use of voice calls for communication during large-scale events. Another profile is composed by users who use only SMS messages to communicate. There is a third group of users who make phone calls and also send SMS. Such user profiles are considered in all of the following stages of the methodology.

3.2. Step 2 – Defining Geographical and Temporal Constraints

The second step of the methodology consists in defining a geographical region and the time window that delimits, spatially and temporally, the occurrence of the large-scale event. This definition must take into account the event nature, the expected audience size, and the event's venue. Once the region of interest is defined, we identify the carrier's base stations that cover this particular region. The analyses should then focus on cellphone calls and SMS messages being sent through the mobile phone network via the identified base stations.

Since it is necessary to focus on the network usage during a period of time around the event's timeframe, we also adopt the timeline notation, shown in Figure 1 for the case of a soccer match, which was proposed first in [Xavier et al. 2013]. The period marked as event corresponds to the duration of the event itself. The other periods are discussed below.

3.3. Step 3 – Identifying Attendees

In the third step, we identify users who made a cellphone call or sent at least one SMS message in one of the selected carrier's base stations at a period that includes the event. The pre_{match} and $post_{match}$ periods (see Figure 1) are periods respectively just before and just after the beginning and after the end of the event. As during the event the attendees normally pay attention to what is happening, it may cause them to not make many phone calls or send SMS messages during the event. In contrast, we expect that people communicate with each other just before or after the event. Thus, in order to identify users

[1]This is an approximation, as some of those users might not have actually attended the event (i.e., event worker or passer-by).

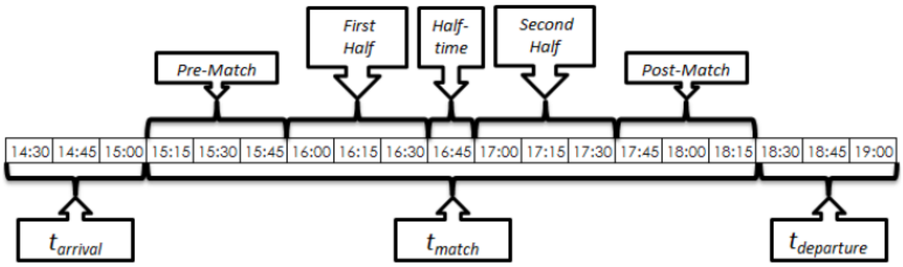

Figure 1. Timeline for workload analysis during an event

that were attending the event, here called *attendees*, we consider the total period from the beginning of pre_{match} until the end of $post_{match}$, here called t_{match}. Users who made at least one cellphone call or sent at least one SMS message during t_{match} are considered *attendees*, i.e., are a target in the conducted analyses.

3.4. Step 4 – Identifying the Movement of Attendees

In this step of the methodology, we identify the subset of attendees who also made cellphone calls or sent SMS messages before or after the soccer match, i.e. before or after the t_{match} period. This identification allows us to track the movement of these users before and after the match. To track these attendees we define two other periods: $t_{arrival}$ and $t_{departure}$ during which people are moving towards the region where the event is happening and leaving it to go somewhere else, respectively. Using the geographical location of the base stations where the voice calls were made or where the SMS messages were sent, it is possible to determine where the attendees came from and where they went to after the event. It is therefore possible to study the workload dynamics in the base stations located on the main access routes to the region where the large-scale event takes place.

For large-scale events, such as the FIFA World Cup and the Olympic Games, these main access ways possibly connect stadiums to hotel regions, to restaurants, and to airports. The identification of these routes can be exploited by many services, such as advertising or public information services.

3.5. Step 5 – Analyzing the Workload

The analysis of the workload imposed by attendees on base stations located within the region of the event, identified according to the previous four steps, focuses on the following aspects:

1. Number of distinct users and distinct attendees
2. Average number of phone calls per distinct user and per distinct attendees
3. Average number of SMS messages per distinct user and per distinct attendees
4. Average time interval between calls and SMS messages, per user and per attendees
5. IAT (Inter Arrival Time) of messages and calls at each base station.

The analysis of time intervals between successive transmissions (IATs) has great value to investigate how the phone call throughput and how transmission of SMS messages occurred. In turn, this analysis is also very important for carriers because an overload on a base station may bring degradation of service quality and consequently an interruption of service (e.g., blocking of phone calls), which could incur financial losses and negative marketing. The analysis of SMS data, including total messages sent by users and the time intervals between transmissions by user and at each base station, along with the definition of usage profiles, can be cited as one key contributions of this work in comparison with [Xavier et al. 2013]. The mobile operator does not provide a log of drop calls. Therefore, we are unable to evaluate the system capacity.

The aforementioned aspects are analyzed not only as aggregate measures computed over the entire t_{match} period, but also at a finer granularity, measured for configurable successive time intervals of 5, 15, 30, 45, and 60 minutes during that period. Thus, workload variation during these intervals may be analyzed. To analyze the workload dynamics in the base stations located near the main access ways to the event, we used heat maps to represent the utilization rates of each base station by mobile phone services and we used these maps to represent the user movement to event venues.

The maps were generated using a function from GoogleMaps Javascript API V3, which takes latitude, longitude, and the communication calls for every analyzed base station as inputs. The region in which a base station is located is represented by a color, indicating the number of calls that went through the base station during the analyzed time span, according to the following color pattern: the darker the color, the larger the number of calls. Thus, it is possible to identify the main access ways to the event analyzing the generated maps for the $t_{arrival}$, t_{match} and $t_{departure}$ intervals.

4. Results

This section shows characterization results and is organized as follows. Subsection 4.1 describes how we applied the proposed methodology in our datasets. Subsection 4.2 characterizes user distribution and Subsection 4.3 presents communication events using timeline notation.

4.1. Applying the Proposed Methodology

According to the methodology described in Section 3, the first step is to define the user's profiles. As we only have data about phone calls and SMS messages, we defined one profile for each case (calls and SMS) and a third profile which combined both of them. The second step consists in defining the event region, and consequently the carrier's base stations that cover this region, as well as defining the time span for the analysis. For the selected soccer matches, the event region was defined as the geographical area covering a radius of 1,500 meters around the Maracanã stadium, the stadium where the matchs took place. In this case, six base stations located within this area were selected for the analysis.

Regarding the time span considered in the analysis, the total time window was the interval between 2:30 PM and 7:15 PM for the 2014 FIFA World Cup matches and the interval between 6:00 PM and 00:00 AM for the Final Match of 2013 Brazilian Cup. The total event duration, including the half-time break, is 105 minutes. We set $t_{arrival}$ and pre_{match} equal to 45 minutes each, whereas a value of 45 minutes was chosen for $t_{departure}$ and $post_{match}$ periods.

4.2. Characterizing the User Profiles

Our dataset was built from anonymized data collected by a major carrier of Brazilian cell in three important matches at the Maracana stadium. The data from Brazilian Cup final

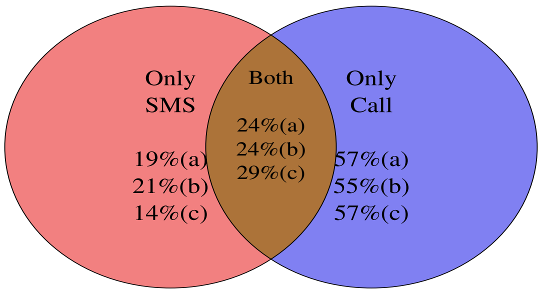

Figure 2. Fraction of identified attendees per profile.

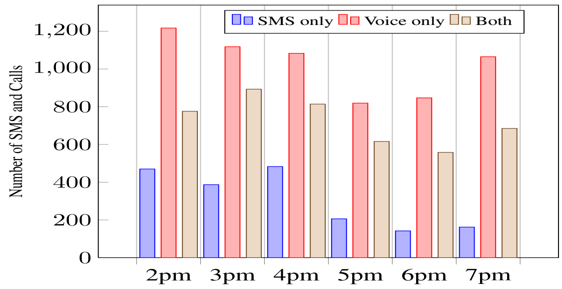

Figure 3. Profiles Timeline - Germany x Argentina

Table 1. User characterization

Profiles			SMS only		Voice only		Both	
Event	Data	Attendance	Users	% Attendees	Users	% Attendees	Users	% Attendees
Flamengo X Atlético-PR	27/Nov/13	57.991	1.428	59,17	5.830	43,57	1.362	80,10
Colombia X Uruguay	18/Jun/14	73.804	983	67,34	2.568	67,68	1.001	75,92
Germany X Argentina	13/Jul/14	74.738	1.220	34,92	3.470	48,10	1.313	64,58

match, Flamengo versus Atletico-PR was colected in November, 27, 2013. The data from World Cup Final Match, Germany versus Argentina, was colected in July, 13, 2014. We also used data from a match between Colombia versus Uruguay at World Cup, colected in June, 18, 2014. As defined, attendees are users who made at least one phone call or sent at least one SMS message during t_{match}. Figure 2 presents the percentages of identified attendees per usage profile, at each match: FLAxATL(a), COLxURU(b) and GERxARG(c). The data network characterization is out of the scope of our work because no dataset is available.

Table 1 presents the number of users and the percentage of attendees in each profile, for the three analyzed soccer matches. We note that the percentages of attendees, in SMS only and Both profiles, are smaller in the World Cup Final Match (GERxARG) than in the Brazilian Cup Final Match (FLAxATL), which indicates a more prevalent presence of users who made phone calls during the entire analyzed time span but not during the t_{match} interval. Such difference occurs in spite of a much larger attendance rate in the former. This can be explained by higher importance and the larger presence of foreigners in matches of the 2014 World Cup. According to FIFA, only 12,984 tickets were purchased by Brazilian residents (around 16% of the stadium's capacity).

4.3. Communication Events

This Section analyzes the number of phone calls and SMS messages sent by attendees during the t_{match} period, focusing on the base stations located within the selected region. Notice that, in Figure 3, we show how the numbers of communication events evolve over time (according to the identified profiles) in intervals of 60 minutes, for the final match between Germany and Argentina. The event took place between 4 and 6 PM on July 13, 2014.

Table 2 show the results for each of the proposed time intervals in the timeline ($t_{arrival}$, pre_{match}, $_{match}$, $post_{match}$, and $t_{departure}$). Overall, as we might expect, the results show that there is a decrease in the number of phone calls and SMS messages

Figure 4. Heat Map - Voice

Figure 5. Heat Map - SMS

Figure 6. Heat Map - SMS and Voice

during the match. The amounts of these events during the $t_{arrival}$ and pre_{match} periods are similar to those of $post_{match}$ and $t_{departure}$ periods.

Table 2. User activity in the timeline – GER x ARG

Profiles	SMS only		Voice only		Both	
Timeline item	SMS	Users	Calls	Users	SMS and Calls	Users
t-arrival	876	423	2.422	1451	1.724	734
pre-match	464	235	1.004	666	760	420
first-half	145	76	440	318	400	238
halftime	61	36	379	294	215	150
second-half	100	51	567	388	391	220
pos-match	102	64	662	473	431	264
t-departure	996	477	2.571	1.471	2.104	794

We also used heat maps to analyze the mobility of attendees at each usage profile, and thus infer the most used routes to get to the event region. Figure 4 show the heat maps, considering all periods from timeline, of "Voice only" profile. In Figure 5, the heat map for the "SMS only" profile, also from all periods from timeline, is shown. Figure 6 exhibit the heat maps for the profile representing a combination of both technologies. We can see that the presence of voice calls is important for mobility studies , once we detected a small number of users at the "SMS only" profile who has mobility during events in timeline. At the match between Germany and Argentina, we observe many attendees coming from Santos Dumont Airport, specially if voice calls are used in order to infer the route.

This characterization pointed out that voice calls and SMS work relatively well in these large-scale events. As both transmission and reception of SMSes tend to remain in operation even during network overload, we propose in next Section an SMS-based platform for transferring small data packets when the data traffic over mobile data networks is unavailable or under congestion.

5. An SMS-based Platform for Transferring Small Data Packages

The Short Message Service (SMS) works in all cellphones. It allows users to send and receive short messages containing up to 160 characters. It can be used for exchanging messages between two people or to deliver the same message to a large number of people. We note that the price of this service is typically fairly low and its availability is relatively high. Indeed, the Brazilian mobile operators are including unlimited SMS plans, despite recent changes in their data plans, which impose restrictions for users to navigate at low speeds. User must purchase an upgrade of their contract or will continue using only voice calls or SMS.

Our SMS-based platform, introduced in this section, will allow the transfer of small data packets also in these situations. In the following, we first present an overview

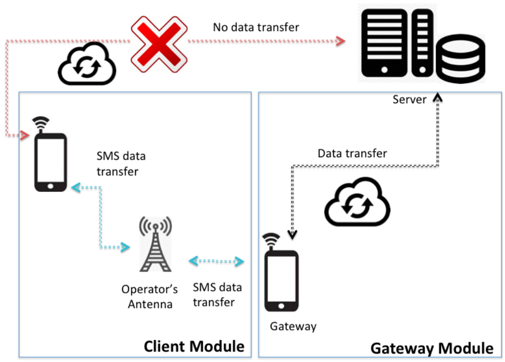

Figure 7. Architecture

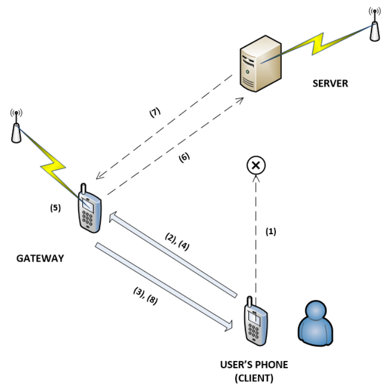

Figure 8. Operation Diagram

of our platform (Section 5.1) and then introduce its general aspects related to its implementation (Section 5.2).

5.1. Description

The main goal of the proposed platform is to keep the traffic of small data packets active even when data networks are not available or are under congestion. Thus, applications that depend on the exchange of such small data packets, running on top of our platform, would continue working. Even if the traffic resources on the data network are scalable, we know they are finite. Thus, the platform provides another communication channel, that can take care of some requests.

The trigger to use our SMS platform must be defined by these application developers. We define as *small*, packets containing up to 5 KBytes. Our platform can transmit this amount of data using approximately just 50 SMSes.

We design the platform carefully to decrease the amount of work needed to adapt existing applications. A common way to transmit data over the Internet is through mecanisms that use POST and GET calls over HTTP, as REST calls. These calls usually send and receive data using a JSON format. We maintain this format of sending and receiving data in our platform. Therefore, an application developer could test if Internet is available. If not, the developer could call our platform just as it would do for common transmission methods.

A gateway acts as a proxy between the device, which cannot access data network, and a service provider. Communication between the device and the gateway is made through a set of SMS. Communication between the gateway and the service server is made through HTTP over a Wi-fi connection. We choose to design the gateway for execution on any Android mobile device, thus allowing the use of entire platform by any individual and removing the need of third party software or hardware. Any kind of application that uses small data packets, such as instant message or news apps could use our approach.

5.2. Implementation

The platform for small data packet transfer was developed in Java for Android platform and is divided into two modules, as shown in Figure 7. The first one, Client Module is a set of methods that allow developers to perform the data transfer via SMS in a totally transparent way, just like they would usually do for transferring data to an specific URL. The main difference in this case is the presence of a cell phone number that will act as a gateway.

The second module, or Gateway Module, is also a java application that must run on a mobile device with access to a mobile phone network to receive SMS. This device must also be connected to a wireless network, in order to transfer the information to the server addressed by the developer of the call procedures in Client Module. When the data network is unavailable, the applications that are running on the device of the user can use our platform to continue communicating with web servers. Figure 8 depicts a workflow illustrating how our platform operates. Since we are proposing a generic platform, i.e. it will be useful for any application that requires an Internet connection, we refer to a requesting application as client. The dotted arrows represent data traffic using HTTP protocols, and the solid arrows represent data traffic using SMS.

Figure 8 shows the eight steps of the operation of the proposed platform. At step (1), the client attempts to communicate with the remote server, but is not able to establish the connection. At this moment, the client asks the user for permission to continue the interaction through SMS. With permission granted, in step (2) the client calls our platform, which in turn, sends an SMS to the gateway and the connection is established between them. In (3), the gateway sends an SMS back to the client as an acknowledgement. Once the connection is established, the client uses the sendAndReceive() method whose parameter and return type are JSON objects. Thus, the platform splits the JSON object into multiple smaller packets and sends them in multiple SMS to the gateway.

In step (5), the gateway rebuilds the JSON object received via SMS. After rebuilding the JSON object, the gateway sends it through a POST to the desired Server in step (6). This step is similar to step (1), however, the gateway is using a network connection available. In step (7), the server returns the result of the POST to the gateway. In step (8), the gateway relays the POST to the platform via a sequence of SMS. The first SMS contains the result of the request and the amount i SMSes it should receive until the result of the POST can be rebuilt (like GET request). After the reconstruction, the JSON object is returned to the client.

We create three types of SMSes, namely CONNECT, ACK and DATA, which have different roles in the communication. Figures 9 and 10 show these SMS types in use by our platform. When a CONNECT SMS is sent by client module, the gateway responds with an ACK SMS, allowing the client to send a defined number of DATA SMS. The data portion of this SMS is Base64 encoded. When all the DATA SMS are received by the gateway module, a complete HTTP POST is done. The structure of each type of SMS is described below:

- CONNECT: CONNECT # Process ID # IMEI # SMS Number # URL
- ACK: ACK # Process ID # Gateway Number
- DATA: DATA # Process ID # Sequential Number # Data Portion

The user must inform the gateway number at the client module. When the gateway module receives the first SMS, it parses the SMS and extracts the origin number (SMS Number).

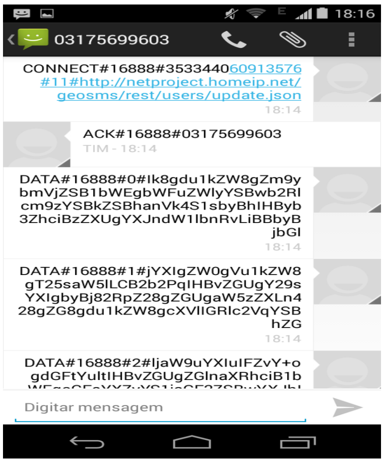

Figure 9. SMS Sequence 1

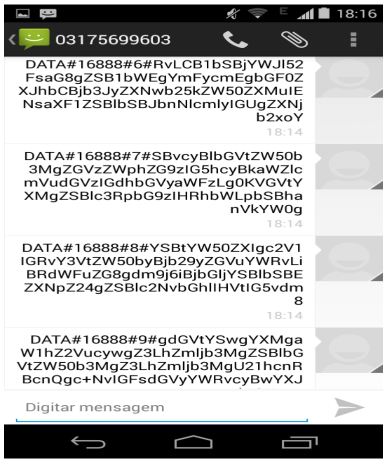

Figure 10. SMS Sequence 2

6. Sample Tool

We developed a sample mobile application in order to validate our SMS-based platform for transferring small data packages.

This tool, SMS and Calls Maps, allows the user to view their history of messages and voice calls marked geographically by latitude and longitude plus time of send/reception on a map, where the user can directly see the message details and content. The user is also able to filter messages and calls by sending regions, perform local search and view usage statistics. These actions can be performed at mobile or at website. In order to see the data on the site is necessary to perform a synchronization step on mobile device. This synchronization originally runs under data network connection, but we also included the SMS-based platform as a synchronize option.

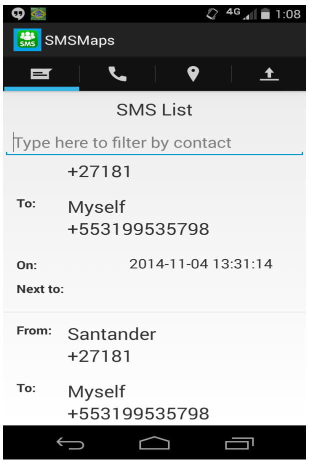

Figure 11. SMS List

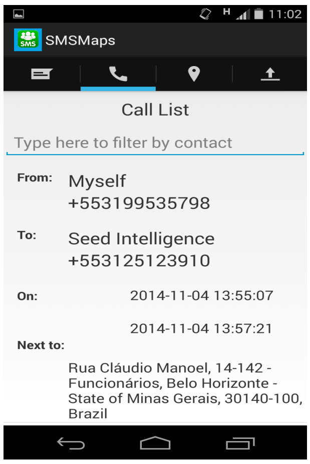

Figure 12. Call LIst

The large number of SMS and Voice Calls that a person may send daily may raise problems related to lack of control and knowledge about old messages that may have been deleted and forgotten by users. Mobile phones provide a list of messages and calls, as seen on Figure 11 and Figure 12, including details, such as date, time and destination number. Yet, instead of simply a list of SMSes and calls, a richer and more meaningful visualization for the user would provide the calls and SMS the user sent and received geographically distributed on a map.

SMS and Call Maps was developed in Java in order to run on mobile devices using Android operating system. The application intercepts all call or SMS events the user sends or receives. In order to allow this interception, the cellphone user must allow specific permissions to our application. These permissions can vary according to the version of the Android operating system installed on phone.

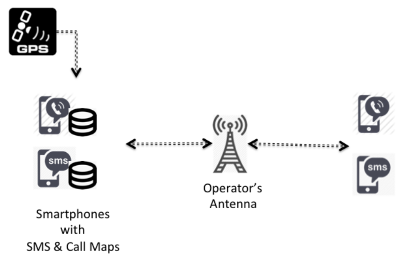

Figure 13. SMS And Call Maps Architecture

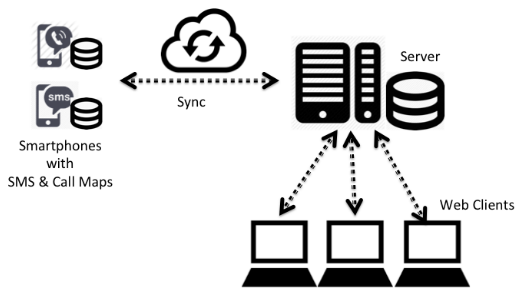

Figure 14. SMS And Call Maps - Sync

Figure 13 shows the architeture of SMS and Call Maps. Geographic coordinates obtained from the GPS running on the cellphone are stored along with the mobile number that originated or received the event as well as the timestamp of the event in a local database. Event timestamp is also stored along with the information mentioned above, in a local database. We need to have access to that information stored in the local database.

We then offer the user the ability to view this information on a website, after it runs a synchronization command in mobile phone. Figures 15 and 16 show screenshots of our application in operation: the first one shows the geographic distribution of all calls and SMS sent/received by the user, whereas the second illustrates a built-in tool that allows the user to view the list of calls and SMS using various filters and export options.

We developed a server module that stores information collected in the mobile devices. This module has been coded in PHP and uses MySQL database. The application allows the user to send the collected data to the server when the mobile device is on the data network coverage. The communication between the android application and PHP server was performed using POST calls, via an API Restfull, as illustrated in Figure 14.

The synchronization between the application and the server module is done over an Internet conection. If this connection is unavailable we use our SMS-based platform to transfer small data packets. Thus, the sync step on figure 14 is replaced by the workflow presented on Figure 8.

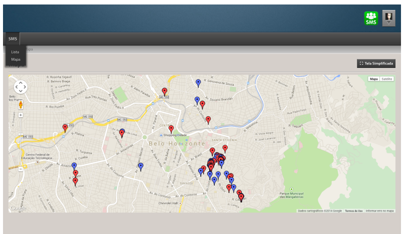

Figure 15. Server Maps

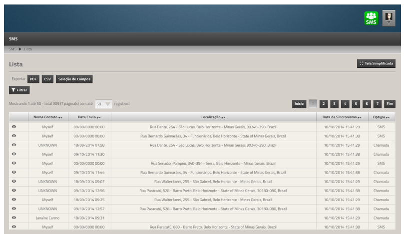

Figure 16. Server Lists

7. Conclusion

This work presents improvements upon the characterization methodology previously proposed by [Xavier et al. 2013]. We included an SMS analysis with the goal of adapting the proposed methodology in order to compare user behavior when using different mobile communication technologies. We analyzed three diferent usage profiles: (i) attendees who made cellphone calls and sent SMS messages, (ii) attendees who only made calls, and (iii) attendees who only sent SMS.

With the analysis of SMS messages and phone calls during soccer matches with high attendance rates, it was possible to identify mobile user behavior patterns when attending such events. We observed that behavior is dependent upon the type of event and usage profile of the mobile device. Voice still prevails as the main means of communication among attendees of these events. The characterization results encouraged us to propose a platform to enrich SMS usage. We present a platform that allows small data packets traffic using SMS. Our main contribution is to keep mobile applications in operation even when the data network is unavailable or congested. We use a cell phone that has Internet access as a gateway. This gateway processes an SMS flow as a request to a web service and returns the results to mobile application also using SMS.

We developed a sample tool in order to validate the proposed platform. SMS and Calls Maps uses a sequence of SMSes to exchange data between mobile application and web server. As future work we expect to share the platform to several developers of mobile applications who rely on the data traffic to keep their applications running when the data network is unavailable or congested. We will also evaluate how to use compression mechanisms along with encryption algorithms.

Acknowledgements

Authors acknowledge CAPES, CNPq, FAPEMIG, FAPERJ, INCTWeb (MCT/CNPq 573871/2008-6) and FIP PUC Minas for their support.

References

Becker, R., Cáceres, R., Hanson, K., Isaacman, S., Loh, J. M., Martonosi, M., Rowland, J., Urbanek, S., Varshavsky, A., and Volinsky, C. (2013). Human mobility characterization from cellular network data. *Commun. ACM*, 56(1):74–82.

Candia, J., González, M. C., Wang, P., Schoenharl, T., Madey, G., and Barabási, A.-L. (2008). Uncovering individual and collective human dynamics from mobile phone records. *Journal of Physics A: Mathematical and Theoretical*, 41(22):224015.

Chen, J., Linn, B., and Subramanian, L. (2009). Sms-based contextual web search. In *Proceedings of the 1st ACM workshop on Networking, systems, and applications for mobile handhelds*, MobiHeld '09, pages 19–24, New York, NY, USA. ACM.

Cheung, M. H. and Huang, J. (2013). Optimal delayed wi-fi offloading. In *Modeling & Optimization in Mobile, Ad Hoc & Wireless Networks (WiOpt), 2013 11th International Symposium on*, pages 564–571. IEEE.

Curto, H. N., Ziviani, A., Caetano, J. A., Malab, C. H. S., Almeida, J. M., and Marques-Neto, H. T. (2014). Characterizing the workload dynamics of a mobile phone network during the fifa 2014 world cup. In *Performance Computing and Communications Conference (IPCCC), 2014 IEEE International*, pages 1–2. IEEE.

Eagle, N. and Pentland, A. (2006). Reality mining: sensing complex social systems. *Personal and ubiquitous computing*, 10(4):255–268.

Eagle, N. and Pentland, A. S. (2009). Eigenbehaviors: Identifying structure in routine. *Behavioral Ecology and Sociobiology*, 63(7):1057–1066.

Estadao, G. (2013). Estadios da copa terao wi-fi gratis. Website. (2013, Jul 23).

Fjeldsoe, B. S., Marshall, A. L., and Miller, Y. D. (2009). Behavior change interventions delivered by mobile telephone short-message service. *American journal of preventive medicine*, 36(2):165–173.

Gonzalez, M. C., Hidalgo, C. A., and Barabasi, A.-L. (2008). Understanding individual human mobility patterns. *Nature*, 453(7196):779–782.

Lee, K., Hong, S., Kim, S. J., Rhee, I., and Chong, S. (2009). Slaw: A new mobility model for human walks. In *INFOCOM 2009, IEEE*, pages 855–863. IEEE.

Oliver, E. (2010). Characterizing the transport behaviour of the short message service. In *Proceedings of the 8th international conference on Mobile systems, applications, and services*, MobiSys '10, pages 223–238, New York, NY, USA. ACM.

Ratti, C., Williams, S., Frenchman, D., and Pulselli, R. (2006). Mobile landscapes: using location data from cell phones for urban analysis. *Environment and Planning b Planning and Design*, 33(5):727.

Reda, A., Noble, B., and Haile, Y. (2010). Distributing private data in challenged network environments. In *Proceedings of the 19th international conference on World wide web*, pages 801–810. ACM.

Soper, D. (2012). Is human mobility tracking a good idea? *Commun. ACM*, 55(4):35–37.

Veeraraghavan, R., Yasodhar, N., and Toyama, K. (2007). Warana unwired: Replacing pcs with mobile phones in a rural sugarcane cooperative. In *ICTD 2007. International Conference*, pages 1–10. IEEE.

Xavier, F., Malab, C., Silveira, L., Ziviani, A., Almeida, J., and Marques-Neto, H. (2013). Understanding human mobility due to large-scale events. In *Third International Conference on the Analysis of Mobile Phone Datasets (NetMob)*.

Zhuo, X., Gao, W., Cao, G., and Dai, Y. (2011). Win-coupon: An incentive framework for 3g traffic offloading. In *Network Protocols (ICNP), 2011 19th IEEE International Conference on*, pages 206–215. IEEE.